

ShiftGuard10: Robust Image Classification

EE708: Fundamentals of Data Science and Machine Intelligence

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Abstract—This project addresses a multi-class image classification problem for the ShiftGuard10 dataset using a deep learning approach. A Residual Neural Network (ResNet-style CNN) is employed to handle complex feature extraction while mitigating vanishing gradient issues. The dataset exhibits class imbalance; therefore, Macro F1-score is used for fair evaluation across classes. Various preprocessing and training strategies are applied to improve generalization and robustness.

I. INTRODUCTION

Image classification is a fundamental problem in computer vision. This work focuses on classifying 32×32 RGB images into 10 classes. The key challenges include limited image resolution and class imbalance. To address these, a deep convolutional architecture along with appropriate preprocessing and evaluation strategies is adopted.

II. DATASET

The dataset consists of 29,400 labeled training images and 7,600 unlabeled test images across 10 classes. The dataset is imbalanced, motivating the use of Macro F1-score, which treats all classes equally.

III. DATA PREPROCESSING

A. Validation Split

The dataset is divided into 90% training and 10% validation sets. The validation set is essential for:

- Monitoring generalization performance during training
- Preventing overfitting by tuning hyperparameters
- Selecting the best model based on validation metrics

B. Data Augmentation

Data augmentation is applied to improve model generalization by artificially increasing data diversity. The transformations include:

- Random cropping with padding: helps the model learn translation-invariant features.
- Horizontal flipping: captures mirror variations in objects
- Color jitter: improves robustness to lighting variations
- Rotation: helps learn orientation-invariant features
- Random erasing: forces the model to rely on distributed features rather than specific regions

These augmentations reduce overfitting and improve robustness to real-world variations.

C. Normalization

Pixel values are normalized using:

$$\mu = (0.4914, 0.4822, 0.4465), \quad \sigma = (0.2470, 0.2435, 0.2616)$$

These values correspond to the dataset-specific mean and standard deviation computed from the training data. Normalization ensures:

- Faster convergence during training
- Stable gradient updates
- Reduced internal covariate shift

IV. MODEL ARCHITECTURE

A. Why ResNet CNN

A Convolutional Neural Network (CNN) is used for the ShiftGuard10 task as it is well-suited for image classification involving 32×32 RGB images. CNNs effectively capture spatial features such as edges, textures, and patterns through convolutional filters, making them highly suitable for learning representations from image data.

However, as CNNs become deeper, they suffer from vanishing gradient and degradation problems, which make training difficult and limit performance.

To address these limitations, a ResNet-style CNN is used. ResNet introduces residual (skip) connections that allow the network to learn residual mappings instead of direct mappings. This enables gradients to flow directly across layers, leading to more stable training of deeper networks.

The residual learning formulation is:

$$H(x) = F(x) + x \tag{1}$$

This design provides the following advantages:

- Easier optimization of deep networks
- Improved feature reuse
- Better convergence and accuracy

B. Network Design

The architecture consists of:

- **Initial convolution layer:** Extracts low-level features such as edges and textures from raw images.
- **Four residual stages (64, 128, 256, 512 channels):** Learn progressively complex and high-level representations while increasing feature capacity.
- **Downsampling via stride-2 convolutions:** Reduces spatial dimensions to capture global context and improve computational efficiency.

- **Global average pooling:** Converts feature maps into a compact representation, reducing overfitting and model complexity.
- **Fully connected layer for classification:** Maps extracted features to class probabilities for final prediction.

V. TRAINING STRATEGY

A. Loss Function

CrossEntropyLoss is used for multi-class classification as it measures the difference between predicted probabilities and true labels.

B. Optimizer

Stochastic Gradient Descent (SGD) with momentum (0.9) is used:

- Momentum accelerates convergence
- Helps avoid getting stuck in local minima.

C. Learning Rate Scheduling

Cosine annealing is used to gradually decrease the learning rate over epochs. This strategy enables:

- **Large updates initially:** Allows the model to learn quickly and explore the parameter space efficiently in early stages.
- **Smaller updates later:** Facilitates fine-tuning of model parameters, helping the model converge to a better local minimum.
- **Smooth decay:** Avoids abrupt changes in learning rate, leading to more stable and consistent training.

D. Regularization and Retraining Strategies

To improve generalization and prevent overfitting, the following techniques are applied:

- **Dropout (0.3):** Randomly deactivates neurons during training, reducing co-adaptation and improving model robustness.
- **Data augmentation:** Introduces variability in training data, helping the model generalize better to unseen samples.
- **Weight decay (5×10^{-4}):** Penalizes large weights, encouraging simpler models and reducing overfitting.
- **Validation-based model selection:** ensures selection of the model with the best validation performance rather than overfitting to training data.

These strategies collectively enhance model stability, improve generalization, and ensure robust performance on unseen data.

VI. EVALUATION METRICS

The model is evaluated using accuracy and Macro F1-score. Macro F1-score is defined as:

$$F1_{macro} = \frac{1}{K} \sum_{k=1}^K F1_k \quad (2)$$

Macro F1-score is particularly useful for imbalanced datasets as it assigns equal importance to each class, regardless of

their frequency. Unlike accuracy, which can be biased toward majority classes, Macro F1-score evaluates both precision and recall for every class independently and then averages them. This ensures that the model's performance on minority classes is properly reflected, leading to a more reliable and balanced evaluation.

VII. RESULTS

- Final training accuracy: 98.38%, indicating strong fitting on training data
- Best validation Macro F1-score: 0.8248

The model performs well on majority classes but shows reduced recall for minority classes due to data imbalance.

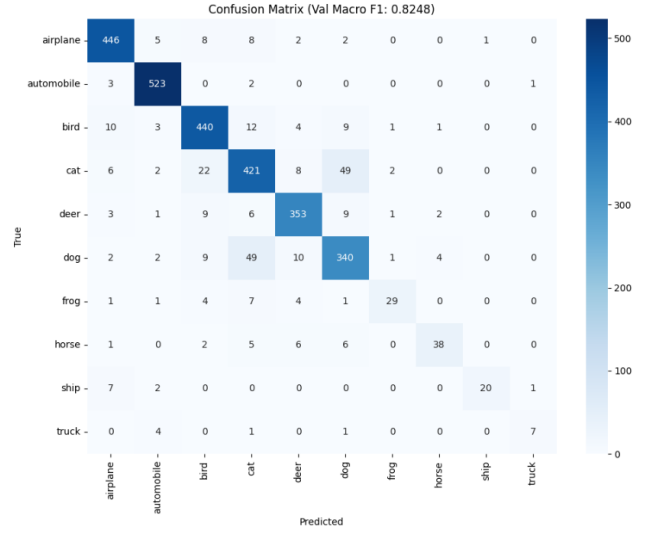


Fig. 1. Confusion matrix on the validation set showing class-wise prediction performance (Macro F1-score: 0.8248)

The confusion matrix shows strong performance on majority classes such as automobile and airplane. However, noticeable confusion exists between visually similar classes like cat and dog, indicating limitations in fine-grained feature discrimination.

VIII. ERROR ANALYSIS

Lower recall in minority classes indicates insufficient representation of these classes in the training data, while confusion between visually similar classes suggests limitations in feature discrimination.

IX. CONCLUSION

- ResNet enables effective deep feature learning
- Data augmentation improves robustness
- Normalization stabilizes training
- Validation strategy helps monitor generalization and select the best model
- Class imbalance impacts performance and requires careful evaluation

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